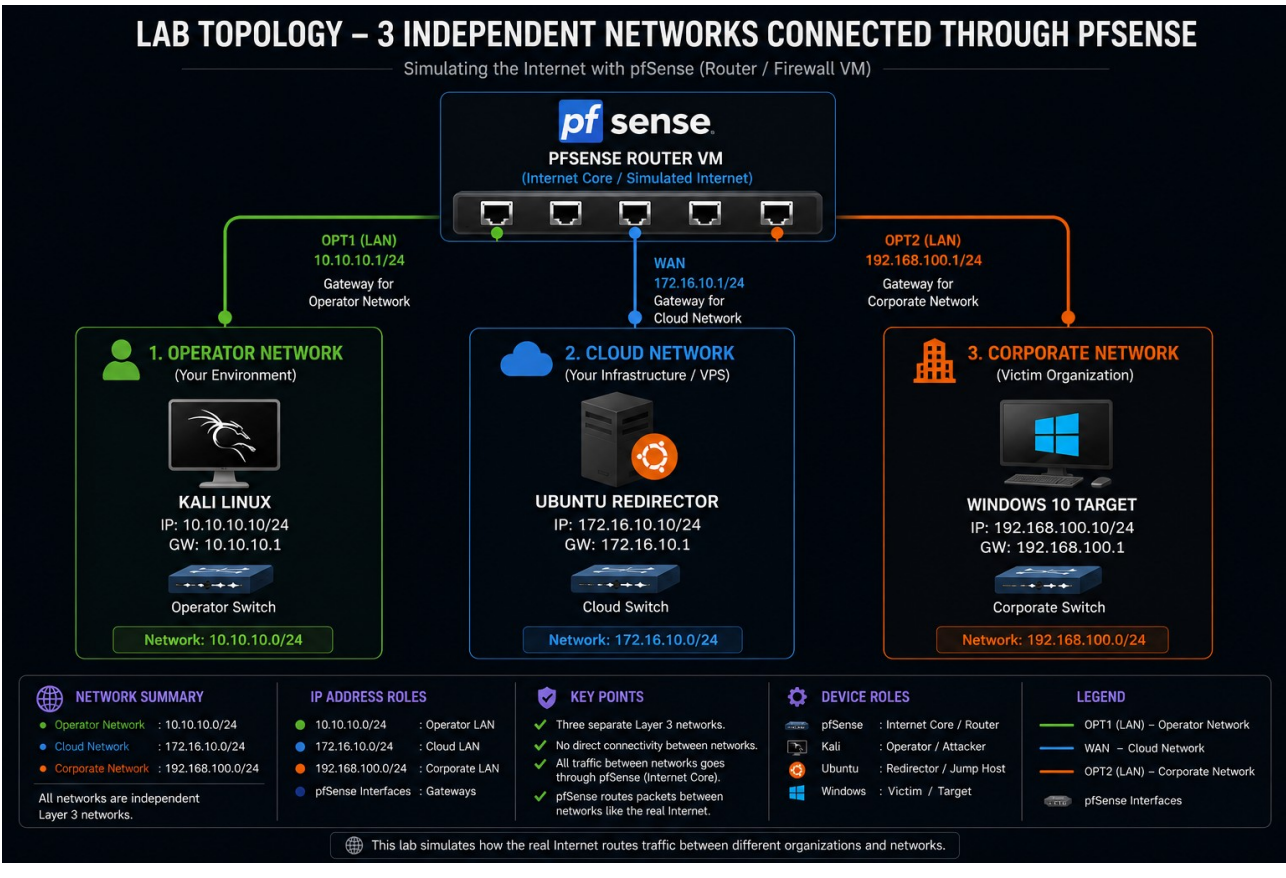


Table of Contents

- 1
- 1
- Red Team Network Architecture & Design.....2
 - Introduction.....2
 - Design Objectives.....3
 - Design Philosophy.....3
 - Network Components.....4
 - PfSense - Internet Core.....4
 - Network A – Operator Network.....4
 - Network B – Cloud Network.....4
 - Network C - Corporate Network.....4
 - Communication Model.....4
 - Security Considerations.....5
 - Real-World Correlation.....5
 - Questions & Answers.....5
- Future Expansion.....5

Red Team Network Architecture & Design



Introduction

What is this document about?

It's about designing a similar realistic environment to practise my offensive security.

A typical Windows post exploitation tutorial mostly begins with the idea that learner already have a prior knowledge of the lab configuration solely focusing on the idea of what's to be done after compromise; without relating it to real life engagement from a Red Team perspective.

Thus, I will be simulating a real life engagement. Whilst I document my learning as a beginner of windows post-exploitation from a red team perspective. Using this document to define the network architecture that underpins the red team laboratory. The environment is intentionally designed to resemble the infrastructure used during real-world engagements rather than a collection of virtual machines sharing a single network. Every design decision is explained from both a networking and operational security perspective so that the laboratory develops both technical understanding and red team trade craft.

Design Objectives

Why does this network architecture exist?

Traditional labs prioritise convenience over realism. This architecture instead prioritises

understanding. The goals are to learn routed communication, Internet-style infrastructure, redirector deployment, packet movement across independent networks, and the **operational reasons** for separating operator infrastructure from target infrastructure. The design also provides a scalable foundation for future command-and-control infrastructure, Active Directory, DNS, logging, VPNs and additional hosts.

Design Philosophy

The lab intentionally separates the operator workstation, redirector, and target workstation into three independent network segments. This design prevents direct communication between the Kali operator and the Windows target. Instead, all command-and-control (C2) traffic is relayed through the Ubuntu Redirector.

The design is intended to:

- understand network segmentation;
- learn SSH tunnelling and traffic forwarding;
- understand the purpose of redirector infrastructure;
- reduce direct exposure of the operator's infrastructure;
- provide a foundation for later pivoting and lateral movement exercises.

Note: This lab focuses on the post-compromise phase of an engagement. Following successful execution of a phishing payload, the Windows target initiates an outbound C2 connection to the Redirector, which relays communications between the target and the operator.

Why isn't every system placed on the same virtual network?

Network Engineering Perspective:

Independent Layer 3 networks expose routing, gateways, packet forwarding, firewall policies and segmentation.

Red Team Perspective:

Operators deliberately compartmentalise infrastructure. The operator, redirector and target should never appear to occupy the same local network because real engagements occur across separate administrative domains connected through Internet routing.

Network Components

PfSense - Internet Core

Why is pfSense the centre of the topology?

Before two independent networks can communicate, a device must know where packets should be forwarded. This responsibility belongs to a router. Within this laboratory, pfSense assumes that role by acting as the default gateway for every network and making forwarding decisions based on the packet's destination.

Network Engineering Perspective:

pfSense performs routing, gateway and firewall functions between the Operator, Cloud and Corporate networks.

Red Team Perspective:

pfSense represents a simplified model of the Internet's routing infrastructure, allowing the lab to reproduce realistic packet paths while remaining manageable.

Network A – Operator Network

Why should the operator never expose their workstation directly?

Network Engineering Perspective:

The Operator Network is an isolated subnet with its own gateway and routing domain.

Red Team Perspective:

During a real engagement, the operator's workstation should never become directly observable by the target environment. Any infrastructure capable of being identified, blocked, or investigated should be considered expendable. For this reason, only the Redirector is intentionally exposed, while the operator's workstation remains isolated behind operational infrastructure.

Network B – Cloud Network

Why is the redirector hosted in its own network?

Network Engineering Perspective: The Ubuntu redirector resides in its own routed network.

Red Team Perspective: The redirector is disposable infrastructure. Targets communicate with it rather than the operator. If it is detected, blocked or 'burned', it can be rebuilt without exposing the operator's trusted environment or backend infrastructure.

Network C - Corporate Network

Why is the target isolated?

Network Engineering Perspective: The Corporate Network is an independent administrative domain with its own addressing and gateway.

Red Team Perspective: It represents the victim environment. Access must be achieved through realistic engagement techniques rather than artificial direct connectivity.

Communication Model

How do packets move during an engagement?

Traffic destined for another network is first sent to the local default gateway. pfSense examines the destination and forwards the packet to the correct network.

Operationally, communications flow through the redirector so that the operator and target never communicate directly, reflecting common red team infrastructure design so as to minimise exposure.

Why does a redirector exist?

The redirector reduces the exposure of the operator, serves as the **publicly reachable endpoint**, supports forwarding to future backend C2 infrastructure, compartmentalises operational assets and enables rapid infrastructure replacement if compromised.

Source	Destination	Expected
Kali	Ubuntu Redirector	Allowed
Ubuntu Redirector	Kali	Allowed
Windows Target	Ubuntu Redirector	Allowed (Outbound C2 Connection)
Ubuntu Redirector	Windows Target	Relayed C2 Traffic

Source
Kali

Destination
Windows Target

Expected
Not Allowed

Security Considerations

Why is infrastructure intentionally segmented?

Segmentation reduces unnecessary trust relationships and allows routing and firewall policies to control traffic.

Also, from an operational perspective it improves OPSEC, limits attribution, reduces blast radius and increases infrastructure resilience.

Real-World Correlation

How does this laboratory map to a real engagement?

- The Operator Network represents the operator's office or home environment.
 - The Cloud Network represents infrastructure hosted by a cloud provider, the Ubuntu system represents an Internet-facing redirector.
 - The Corporate Network represents the target organisation.
 - And pfSense compresses the Internet's routing infrastructure into a single controllable device for learning purposes.
-

Questions & Answers

What is a gateway?

What is type of topology is this?

What are the things the red team operator's are meant to not expose?

Is it only publicly reachable device/node with IP can be used for redirector?

Future Expansion

How will the architecture evolve?

The architecture is designed to support backend C2 servers, multiple redirectors, enterprise firewalls, DNS, Active Directory, VPNs, logging infrastructure, additional operator workstations and multiple corporate subnets without changing the unde